

Data Science Project: Predicting Customer Churn Through Behavioural, Financial and Customer Experience Analytics

Introduction

Customer churn represents a significant challenge for subscription-based organisations as the loss of existing customers can reduce revenue, increase acquisition costs, and negatively affect long-term business performance. In this context, customer churn refers to the proportion of customers who discontinue a product or service within a given period, either by cancelling their subscription, failing to renew a contract, or moving to a competitor. A retained customer is a customer who continues to use the organisation's service and remains subscribed, while a churned customer is a customer who terminates or discontinues the service. Within this dataset, retained customers are represented by a target value of 0, while churned customers are represented by a target value of 1.

The aim of this project was to investigate whether customer demographic, behavioural, financial and customer experience variables could be used to predict customer churn. The dataset consisted of 10,000 customer records and 32 attributes relating to customer demographics, engagement activity, billing behaviour, customer support interactions and satisfaction indicators. Data profiling identified a mixture of categorical and numerical variables suitable for exploratory analysis and predictive modelling.

Three research hypotheses were developed. Firstly, customers exhibiting lower engagement levels would be more likely to churn. Secondly, customers experiencing lower satisfaction levels would demonstrate a greater probability of churn. Finally, customers encountering payment-related issues would be more likely to discontinue their subscription.

To investigate these hypotheses, exploratory data analysis, feature engineering, predictive modelling and dashboard visualisation techniques were applied. Multiple machine learning models were evaluated, including Logistic Regression, Random Forest and a SMOTE-enhanced Random Forest model. Particular attention was given to model evaluation metrics including accuracy, precision, recall and F1-score due to the imbalanced nature of the target variable, where approximately 89.8% of customers were retained and 10.2% had churned. The findings of this project provide insights into the factors influencing customer retention and support evidence-based business recommendations for reducing customer attrition.

Data Infrastructure

A combination of data science, machine learning and business intelligence tools was used throughout this project to support data preparation, analysis, predictive modelling

and stakeholder communication. The selected technology stack provided an end-to-end analytical workflow capable of transforming raw customer data into actionable business insights.

Python was selected as the primary analytical language due to its extensive data science ecosystem and ability to efficiently process large datasets. Pandas and NumPy were used for data manipulation, cleansing and transformation, while Matplotlib and Seaborn were used to create exploratory visualisations that supported the identification of patterns and customer trends. The dataset contained 10,000 customer records and 32 variables, requiring tools capable of handling both numerical and categorical features effectively.

Google Colab was initially used as the development environment as it provided an accessible and reproducible notebook workspace for exploratory analysis, feature engineering and machine learning experimentation. The completed analysis was subsequently maintained within Visual Studio Code to support version control, evidence collection and project organisation.

Scikit-learn was used to develop and evaluate predictive machine learning models, including Logistic Regression and Random Forest. Following the identification of class imbalance within the target variable, the Synthetic Minority Oversampling Technique (SMOTE) was applied to improve churn detection and enhance model evaluation.

Power BI was used to develop an interactive dashboard that communicated customer churn insights to business stakeholders. Dashboard pages focused on customer behaviour, customer experience and churn drivers, enabling analytical findings to be presented in a clear and accessible format.

The selected toolset provided a balance between analytical capability and business usability. Python enabled advanced data preparation and predictive modelling, while Power BI supported effective communication and decision-making through interactive visualisations, ensuring the project addressed both technical and business requirements.

Data Engineering

Data engineering activities were undertaken to prepare the dataset for exploratory analysis and predictive modelling. The dataset consisted of 10,000 customer records and 32 variables covering customer demographics, engagement behaviour, billing activity, customer support interactions and satisfaction metrics. Initial profiling identified a mixture of numerical and categorical variables requiring transformation before machine learning techniques could be applied.

```
# Dataset dimensions
print("Dataset Shape:")
print(df.shape)

print("\n")
print("Column Names:")
print(df.columns.tolist())

*** Dataset Shape:
(10000, 32)

Column Names:
['customer_id', 'gender', 'age', 'country', 'city', 'customer_segment', 'tenure_months', 'signup_channel', 'contract_type', 'monthly_logins', 'weekly_active_days', 'avg_session_time', 'features_used', 'usage_growth_rate', ...]
```

Figure 1. Dataset Structure and Variable Types

A data quality assessment was performed using dataset profiling and missing value analysis. The results showed that the dataset was largely complete, with only one variable, `complaint_type`, containing missing values. A total of 2,045 missing records were identified, while all other attributes were fully populated. Rather than removing these observations, the missing values were retained and treated as customers without a recorded complaint to minimise data loss and preserve business context

```
#Data types and completeness
df.info()

*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 32 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   customer_id           10000 non-null  object 
 1   gender                10000 non-null  object 
 2   age                  10000 non-null  int64  
 3   country               10000 non-null  object 
 4   city                 10000 non-null  object 
 5   customer_segment      10000 non-null  object 
 6   tenure_months         10000 non-null  int64  
 7   signup_channel        10000 non-null  object 
 8   contract_type         10000 non-null  object 
 9   monthly_logins        10000 non-null  int64  
10  weekly_active_days    10000 non-null  int64  
11  avg_session_time      10000 non-null  float64 
12  features_used         10000 non-null  int64  
13  usage_growth_rate     10000 non-null  float64 
14  last_login_days_ago   10000 non-null  int64  
15  monthly_fee           10000 non-null  int64  
16  total_revenue         10000 non-null  object 
17  payment_method        10000 non-null  object 
18  payment_failures      10000 non-null  int64  
19  discount_applied      10000 non-null  object 
20  price_increase_last_3m 10000 non-null  object 
21  support_tickets       10000 non-null  int64  
22  avg_resolution_time    10000 non-null  float64 
23  complaint_type        7955 non-null   object 
24  csat_score            10000 non-null  int64  
25  escalations           10000 non-null  int64  
26  email_open_rate       10000 non-null  float64 
27  marketing_click_rate   10000 non-null  float64 
28  nps_score             10000 non-null  int64  
29  survey_response       10000 non-null  object 
30  referral_count        10000 non-null  int64  
31  churn                 10000 non-null  int64  
dtypes: float64(5), int64(15), object(12)
memory usage: 2.4+ MB
```

Figure 2. Missing Value Analysis Across Dataset Variables

To prepare the data for modelling, categorical variables including gender, country, city, customer segment, contract type and payment method were transformed into machine-readable formats through one-hot encoding. Numerical variables were standardised using the StandardScaler technique prior to Logistic Regression modelling to ensure variables measured on different scales did not disproportionately influence model performance.

Feature engineering was undertaken to create additional business-focused metrics. An Engagement Score was created by combining monthly logins, weekly active days and features used. A Loyalty Score and Support Risk Score were also developed using customer satisfaction, support ticket and escalation variables. Analysis of the engineered features showed that retained customers achieved higher Engagement Scores than churned customers, indicating that customer activity levels may be associated with retention outcomes.

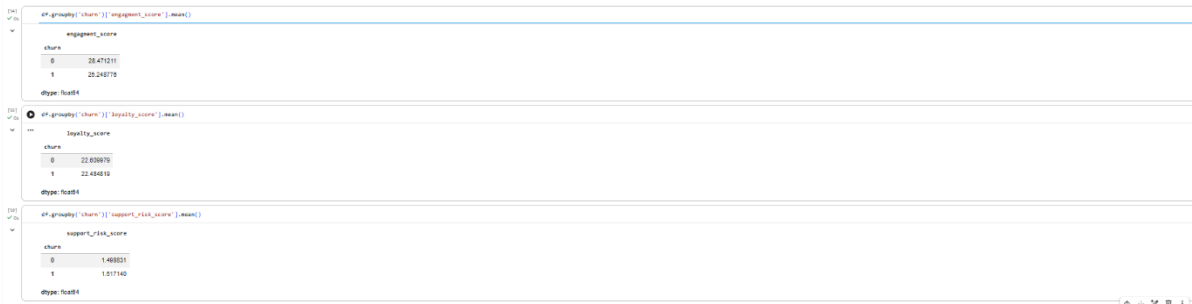


Figure 3. Comparison of Engineered Features by Churn Status

The completed data engineering process improved data quality, enhanced model readiness and generated additional variables that supported subsequent analytical and modelling activities.

Data Visualisation & Dashboard

Data visualisation was used to communicate analytical findings to both technical and non-technical stakeholders. Microsoft Power BI was used to develop an interactive dashboard that transformed analytical outputs into business-focused insights relating to customer retention and churn. The dashboard was organised into four sections covering executive performance, customer behaviour, customer experience and churn drivers.

The Executive Overview dashboard provided a high-level summary of key performance indicators including total customers, churn rate, average customer revenue and average customer tenure. Customer segment and churn distribution visualisations enabled stakeholders to understand the composition of the customer base and the scale of customer attrition.

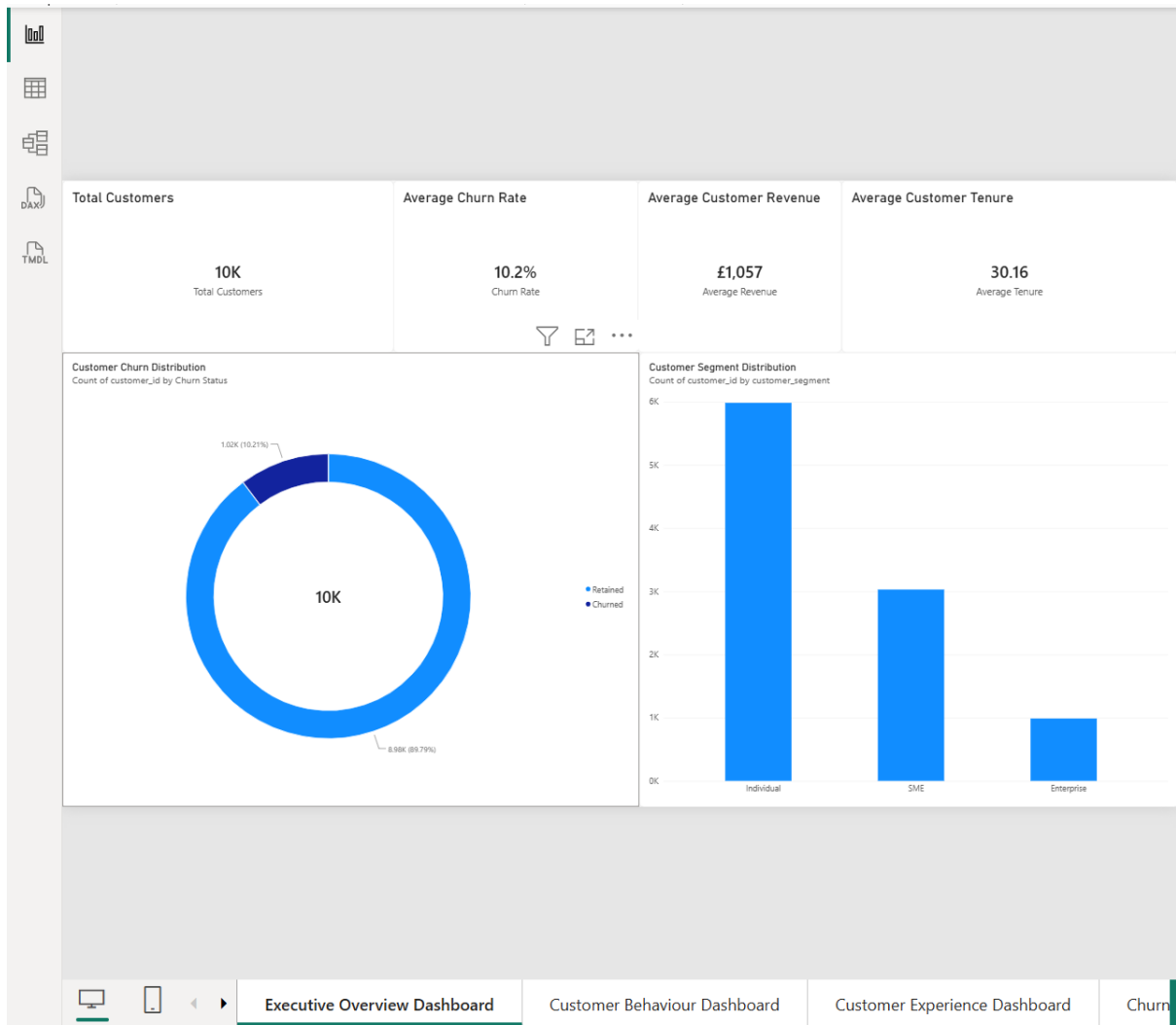


Figure 4. Executive Overview Dashboard

The Customer Behaviour dashboard focused on engagement-related metrics including monthly logins, engagement score, usage growth rate and customer inactivity. These visualisations supported the investigation of behaviour patterns associated with churn.

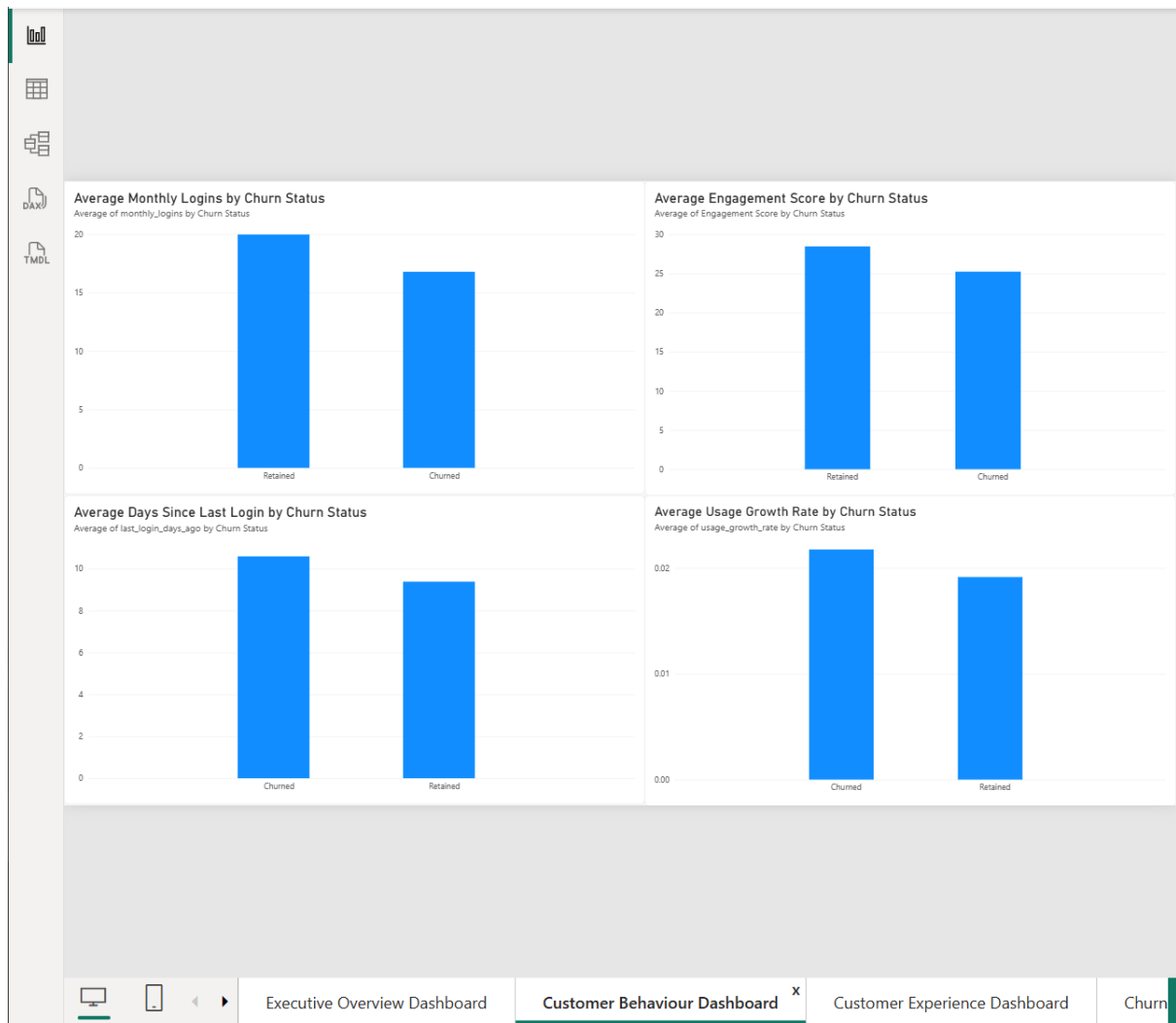


Figure 5. Customer Behaviour Dashboard

The Customer Experience dashboard presented customer satisfaction, payment failures, complaint types and support metrics, allowing service-related drivers of churn to be identified.

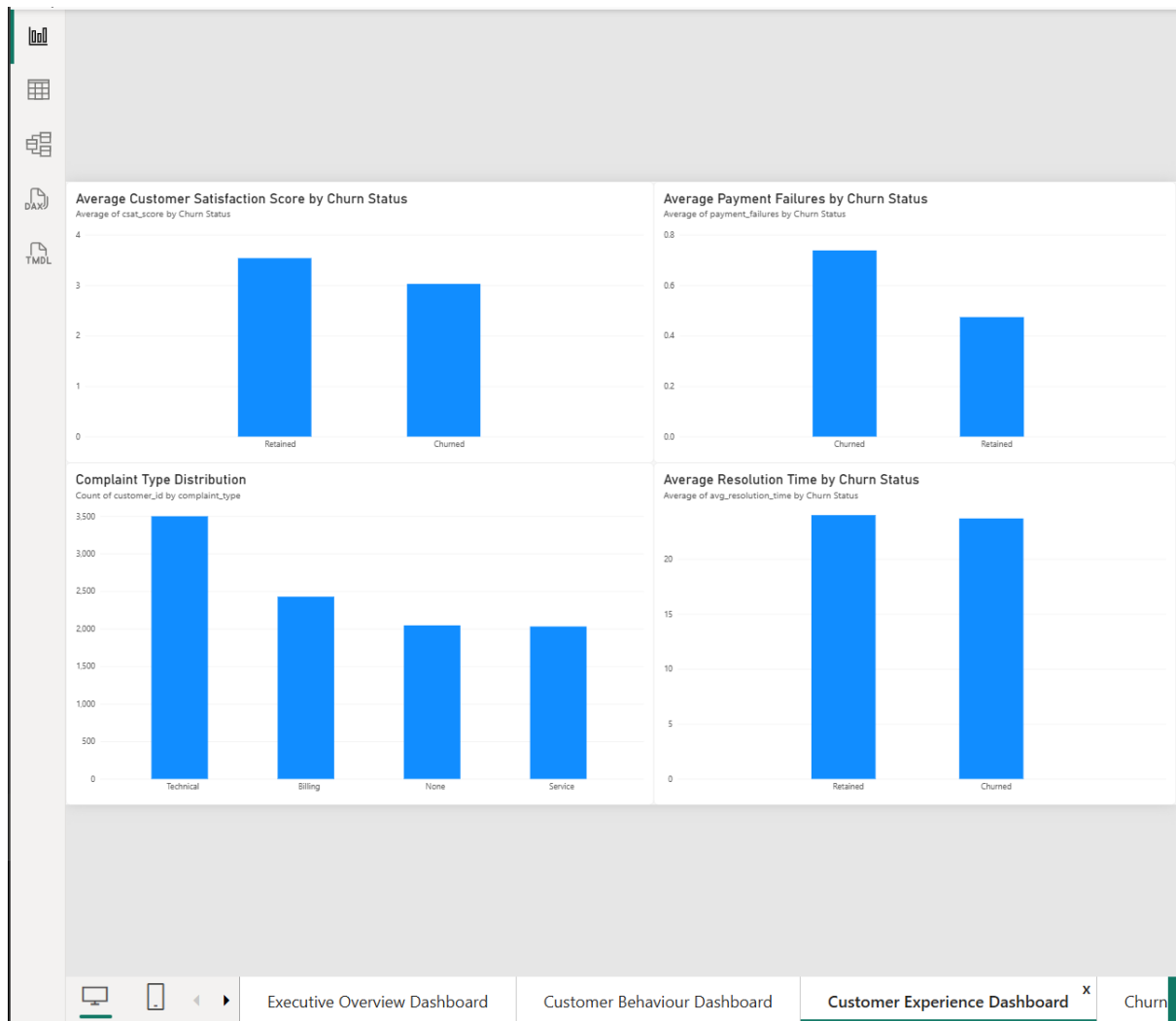


Figure 6. Customer Experience Dashboard

The final dashboard explored churn drivers across customer segments, countries and contract types, while translating findings into actionable business recommendations.

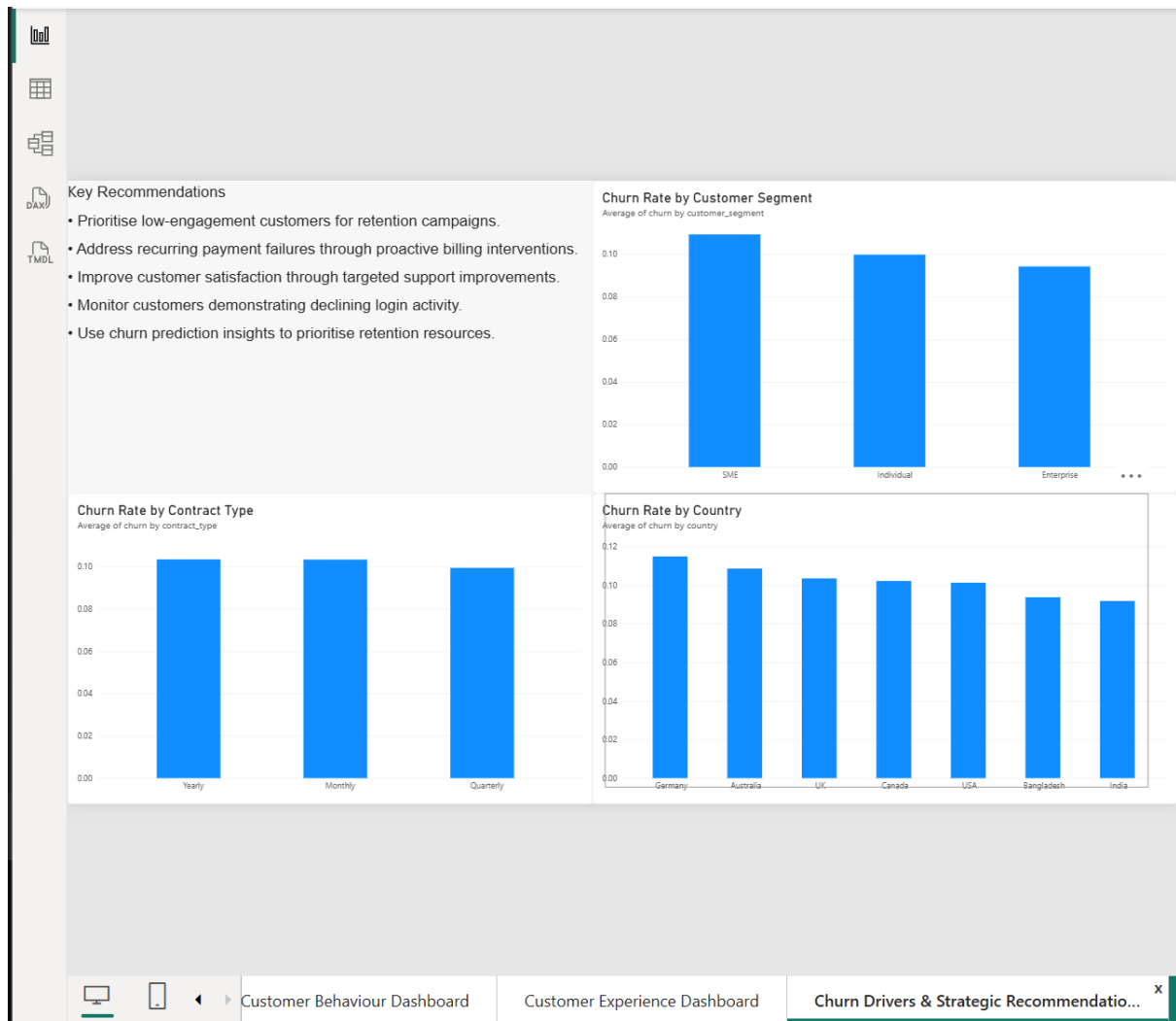


Figure 7. Churn Drivers and Strategic Recommendations Dashboard

Data Analytics

Customer churn analysis was undertaken to identify factors associated with customer attrition and evaluate the effectiveness of predictive modelling techniques. Initial analysis of the target variable identified a significant class imbalance, with approximately 89.8% of customers retained and 10.2% of customers classified as churned. This distribution presented a challenge for predictive modelling because machine learning algorithms had a tendency to favour the majority class, making

accuracy alone an unreliable performance metric

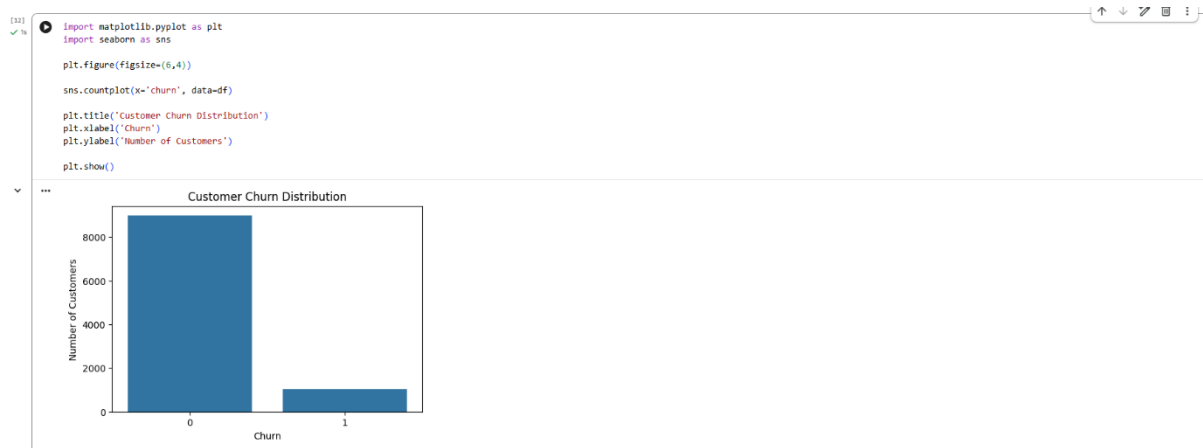


Figure 8. Distribution of Retained and Churned Customers

Exploratory analysis was conducted to investigate the relationship between customer behaviour and churn. Results indicated that retained customers exhibited higher levels of engagement than churned customers, particularly in relation to monthly platform usage. Customers who churned demonstrated lower login frequency and lower engagement scores, suggesting that declining product usage may act as an early warning indicator of future customer attrition (Figure 9). Furthermore, churned customers recorded lower customer satisfaction scores and experienced more payment failures than retained customers, highlighting the potential influence of customer experience and billing-related issues on retention outcomes.

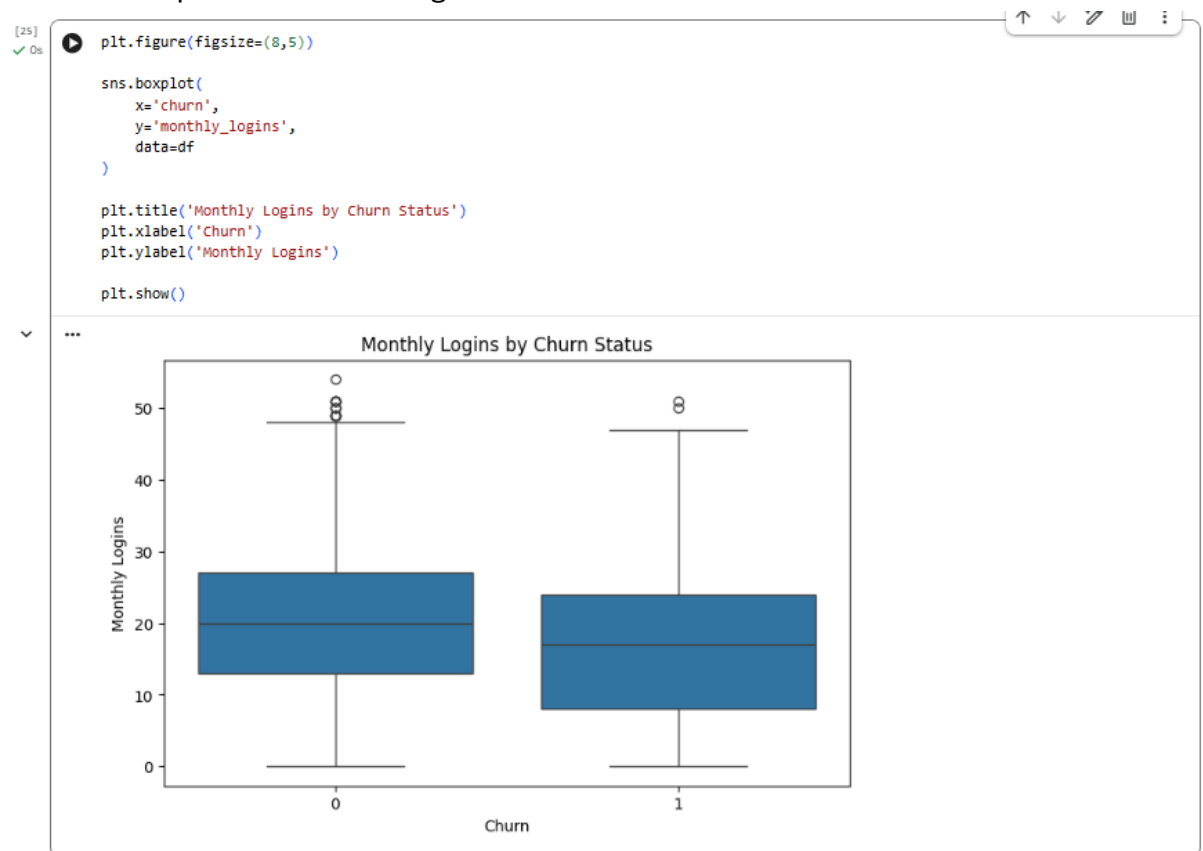


Figure 9. Monthly Login Activity by Churn Status

A Logistic Regression model was selected as the baseline classification model due to its suitability for binary classification problems. Whilst the model achieved an accuracy of 89.65%, the confusion matrix revealed extremely poor recall for churned customers, correctly identifying only two churned customers while misclassifying 202 churned customers as retained. This demonstrates that accuracy alone can provide a misleading representation of model effectiveness when analysing imbalanced datasets.

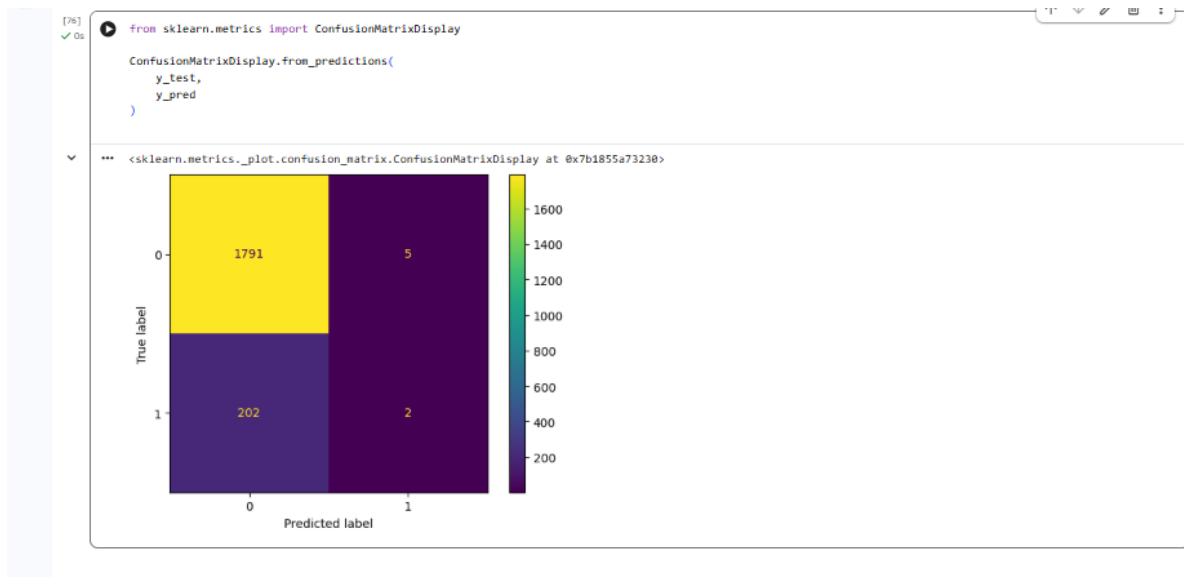


Figure 10. Logistic Regression Confusion Matrix

A Random Forest model and a SMOTE-enhanced Random Forest model were subsequently evaluated. Although SMOTE improved recall and F1-score compared with previous models, overall predictive performance remained limited. This suggests that the available predictors provided only moderate separation between retained and churned customers. However, feature importance analysis identified tenure months, customer satisfaction score, monthly logins and total revenue as the most influential predictors of churn. These variables provide useful insight for future customer retention strategies.

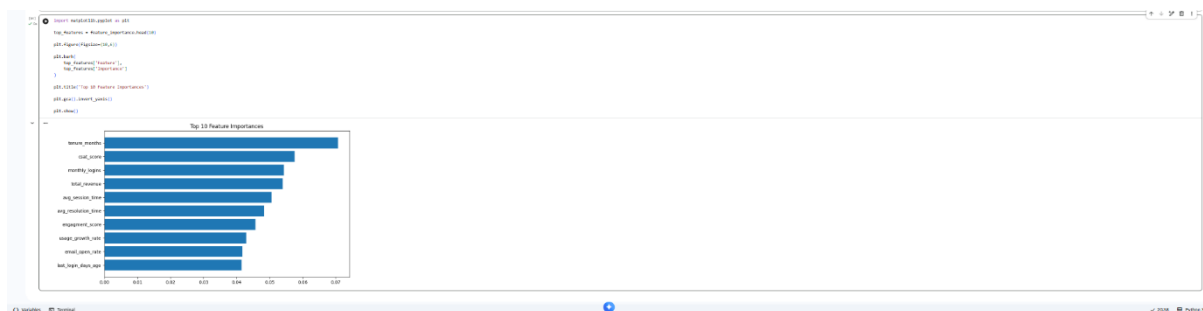


Figure 11. Top 10 Feature Importances Identified by the Random Forest Model

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